Supplementary Material
How Hard Is the Attack? A Complexity-Theoretic Systematization of Machine Learning Security

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1. Classifying the corpus

This section reports the corpus in two layers. Tables 2 to 4 cross-tabulate features against one another, with works in the cells by citation number. Tables 5 and 6 give one row per work, so that every cell above is traceable to the paper behind it. Every value comes from a fixed vocabulary settled before the pass began.

The cross-tabulations are the layer that carries an argument, because in them an empty cell is a combination nobody has published. The value `none` is an answer and not a missing entry, recording that a work constrains no party at all. Of the 181 works classified so far, 32 state no resource bound on any party, and 0 fall outside the frame in the sense that deleting every resource restriction leaves the main claim unchanged.

This is a partial pass. 181 of the corpus's works are classified. The remainder are not yet placed.

TABLE 1: The corpus by pipeline phase (rows) against access level (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

phase	**hard label**	**scores**	**black box**	**white box**	**weights**	**gradients**	**channel**	**none**	***unassigned***	Σ
train time	[55]		[98], [131]	[61], [64], [123], [125], [149], [167], [177], [179]		[25]	[174]	[4], [11], [16], [19], [28], [30], [40], [63], [65], [73], [118], [144], [146], [151], [172], [173]		29
test time	[85]		[26], [50], [89], [90], [94], [142], [150], [157]	[6], [12], [42], [43], [48], [58], [59], [91], [128], [129], [156], [159], [160], [164], [166], [169]			[104]	[9], [34], [37], [44], [71], [78], [86], [101], [122], [127], [135], [136], [140], [153], [181]		41
post hoc		[15], [46], [47], [54], [79], [106], [111], [143], [155], [163], [178]	[45], [67], [70], [102], [113], [137], [154], [170]	[22], [51], [52], [88], [93], [114], [117], [120], [130], [141], [147], [165], [168], [175]	[110]	[115]		[56], [83], [134], [138], [162]	[148]	41
n/a		[72]						[1], [2], [3], [5], [7], [8], [10], [13], [14], [17], [18], [20], [21], [23], [24], [27], [29], [31], [32], [33], [35], [36], [38], [39], [41], [49], [53], [57], [60], [62], [66], [68], [69], [74], [75], [76], [77], [80], [81], [82], [84], [87], [92], [95], [96], [97], [99], [100], [103], [105], [107], [108], [109], [112], [116], [119], [121], [124], [126], [132], [133], [139], [145], [152], [158], [161], [171], [176], [180]		70
Σ	2	12	18	38	1	2	2	105	1	181

TABLE 2: The corpus by constrained party (rows) against resource unit (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies. A dagger marks works where the resource appears only as a side condition rather than as a parameter the work assesses on purpose.

party	time	circuit size	queries	samples	memory	algo class	parameters	perturbation	corruption	rounds	privacy budget	circuit depth	success p
attacker	[10], [23], [27], [39], [55], [56], [64], [67], [72], [74], [85], [93], [99], [106], [107], [114], [124], [132], [133], [163], [167], [174], [25], [47], [88], [157], [177], [179]†	[1], [20], [32], [41], [92], [94]	[5], [36], [45], [50], [111], [130], [46], [54], [89], [137], [155]†	[110], [148]†	[21], [76], [77], [103]	[18], [53], [82], [113], [171]		[6], [12], [43], [58], [59], [71], [86], [101], [104], [122], [127], [135], [136], [140], [153], [156], [159], [160], [164], [166], [169], [42], [48]†	[123], [125], [131], [144], [151]			[100]	
learner	[14], [17], [28], [30], [31], [34], [38], [62], [65], [69], [73], [108], [112], [121], [126], [161], [180]	[96]	[19]†	[9], [44], [63], [66], [68], [75], [81], [128], [129], [172]	[146]	[2], [3], [7], [16], [37], [49], [60], [84], [87], [97], [118], [158], [173]	[40], [95]						
auditor	[51], [61], [98], [102], [117], [26], [178]†			[138], [181]		[29], [70], [105]							
mechanism none	[24], [145]				[80], [119]	[176]†					[4], [147]†		
Σ	54	7	12	14	7	22	2	23	5	0	2	1	0

TABLE 3: The corpus by resource unit (rows) against tightness of the bound (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

unit	**unstated**	**poly**	**quasi-poly**	**class**	**budget**	**conditional**	**exponent**	**measured**	Σ
time	[114], [178]	[24], [31], [34], [51], [55], [56], [61], [65], [72], [73], [74], [85], [93], [98], [99], [102], [106], [107], [108], [112], [117], [126], [132], [133], [161], [163], [167], [174], [180]	[28]		[145], [179]	[10], [14], [17], [30], [38], [62]	[23], [26], [27], [39], [69], [121], [124], [177]	[25], [47], [64], [67], [88], [157]	54
circuit size		[94]	[1], [41], [92]		[32]	[96]	[20]		7
queries					[19], [45], [89], [130], [137]		[5], [36], [50], [111]	[46], [54], [155]	12
samples				[63], [172]	[9], [44], [128], [129], [181]		[66], [68], [75], [81], [110], [138]	[148]	14
memory					[146]	[80]	[21], [76], [77], [103], [119]		7
algo class				[2], [3], [7], [16], [18], [29], [37], [49], [53], [60], [70], [82], [84], [87], [97], [105], [113], [118], [158], [171], [173], [176]					22
parameters					[40], [95]				2
perturbation					[6], [12], [42], [43], [48], [58], [59], [71], [86], [101], [104], [122], [127], [135], [136], [140], [153], [156], [159], [160], [164], [166], [169]				23
corruption					[123], [125], [131], [151]			[144]	5
rounds									0
privacy budget					[4], [147]				2
circuit depth							[100]		1
success prob									0
none	[8], [11], [13], [15], [22], [33], [35], [52], [57], [78], [79], [83], [90], [91], [109], [115], [116], [120], [134], [139], [141], [142], [143], [149], [150], [152], [154], [162], [165], [168], [170], [175]								32
Σ	34	30	4	24	45	8	25	11	181

TABLE 4: The corpus by attack family (rows) against constrained party (columns). Each cell lists every work with that combination, by citation number. An empty cell is a combination no work in the corpus occupies.

family	**attacker**	**learner**	**auditor**	**mechanism**	**none**	Σ
membership	[47], [93], [110], [155], [167]		[178]		[15], [79], [83], [134], [143], [175]	12
attribute						0
reconstruct	[25], [56], [72], [106], [111], [130]				[22], [52], [120], [170]	10
distribution	[163], [174]					2
extract	[45], [46], [54], [67], [113]				[154]	6
invert						0
forge	[64], [77], [88], [114], [179]				[115], [149], [165]	8
evade	[6], [12], [42], [43], [48], [58], [59], [71], [85], [86], [89], [94], [101], [104], [122], [127], [135], [136], [140], [153], [156], [157], [159], [160], [164], [166], [169]	[9], [34], [37], [44], [73], [128], [129]	[26]		[90], [91]	37
evade audit	[50], [177]		[138]		[142], [162]	5
inject			[181]			1
poison	[123], [125], [131], [151]	[19], [63], [65], [172]				8
backdoor	[137], [144]		[51], [61], [98], [102], [117]		[11], [141], [168]	10
multiple					[116], [139]	2
none	[1], [5], [10], [18], [20], [21], [23], [27], [32], [36], [39], [41], [53], [55], [74], [76], [82], [92], [99], [100], [103], [107], [124], [132], [133], [171]	[2], [3], [7], [14], [16], [17], [28], [30], [31], [38], [40], [49], [60], [62], [66], [68], [69], [75], [81], [84], [87], [95], [96], [97], [108], [112], [118], [121], [126], [146], [158], [161], [173], [180]	[29], [70], [105]	[4], [24], [80], [119], [145], [176]	[8], [13], [33], [35], [57], [109], [150], [152]	77
unassigned	[148]			[147]	[78]	3
Σ	85	45	12	7	32	181

TABLE 5: Per-work reference layer: attack scenario and threat model.

#	Target	Family	Phase	Access	Write	Frame
[83]	train data	membership	post hoc	none	read only	none
[143]	train data	membership	post hoc	scores	read only	none
[79]	train data	membership	post hoc	scores	read only	none
[167]	train data	membership	train time	white box	corpus	compute
[110]	train data	membership	post hoc	weights	read only	statistical
[155]	train data	membership	post hoc	scores	read only	compute
[178]	train data	membership	post hoc	scores	read only	compute
[47]	train data	membership	post hoc	scores	read only	compute
[15]	train data	membership	post hoc	scores	read only	none
[175]	train data	membership	post hoc	white box	read only	none
[134]	train data	membership	post hoc	none	none	none
[93]	train data	membership	post hoc	white box	read only	compute
[52]	train data	reconstruct	post hoc	white box	read only	none
[56]	train data	reconstruct	post hoc	none	read only	compute
[72]	train data	reconstruct	n/a	scores	read only	compute
[120]	train data	reconstruct	post hoc	white box	read only	none
[25]	train data	reconstruct	train time	gradients	read only	compute
[111]	train data	reconstruct	post hoc	scores	read only	compute
[130]	train data	reconstruct	post hoc	white box	read only	compute
[22]	train data	reconstruct	post hoc	white box	read only	none
[170]	train data	reconstruct	post hoc	black box	read only	none
[106]	train data	reconstruct	post hoc	scores	read only	compute
[163]	train data	distribution	post hoc	scores	read only	compute
[174]	train data	distribution	train time	channel	corpus	compute
[45]	parameters	extract	post hoc	black box	read only	compute
[154]	parameters	extract	post hoc	black box	read only	none
[67]	parameters	extract	post hoc	black box	read only	compute
[54]	parameters	extract	post hoc	scores	read only	compute
[113]	parameters	extract	post hoc	black box	read only	compute
[46]	parameters	extract	post hoc	scores	read only	compute
[165]	training run	forge	post hoc	white box	corpus	none
[77]	none	forge	n/a	none	none	compute
[115]	train data	forge	post hoc	gradients	corpus	none
[179]	training run	forge	train time	white box	model	compute
[88]	training run	forge	post hoc	white box	corpus	compute
[64]	training run	forge	train time	white box	model	compute
[114]	training run	forge	post hoc	white box	model	compute
[149]	training run	forge	train time	white box	corpus	none
[164]	outputs	evade	test time	white box	read only	capability
[48]	outputs	evade	test time	white box	read only	capability
[153]	outputs	evade	test time	none	read only	capability
[37]	outputs	evade	test time	none	read only	compute
[34]	outputs	evade	test time	none	none	compute
[101]	outputs	evade	test time	none	read only	capability
[86]	outputs	evade	test time	none	read only	capability
[9]	outputs	evade	test time	none	read only	statistical
[122]	outputs	evade	test time	none	read only	capability
[58]	outputs	evade	test time	white box	read only	capability
[136]	outputs	evade	test time	none	read only	capability
[85]	outputs	evade	test time	hard label	read only	compute
[94]	outputs	evade	test time	black box	read only	compute
[127]	outputs	evade	test time	none	read only	capability
[140]	outputs	evade	test time	none	read only	capability
[135]	outputs	evade	test time	none	read only	capability
[26]	outputs	evade	test time	black box	read only	compute
[91]	outputs	evade	test time	white box	read only	none
[90]	outputs	evade	test time	black box	read only	none
[71]	outputs	evade	test time	none	read only	capability

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[59]	outputs	evade	test time	white box	read only	capability
[6]	outputs	evade	test time	white box	read only	capability
[42]	outputs	evade	test time	white box	read only	capability
[44]	outputs	evade	test time	none	read only	statistical
[169]	outputs	evade	test time	white box	read only	capability
[160]	outputs	evade	test time	white box	read only	capability
[156]	outputs	evade	test time	white box	read only	capability
[129]	outputs	evade	test time	white box	read only	statistical
[43]	outputs	evade	test time	white box	read only	capability
[166]	outputs	evade	test time	white box	read only	capability
[104]	outputs	evade	test time	channel	channel	capability
[159]	outputs	evade	test time	white box	read only	capability
[89]	outputs	evade	test time	black box	read only	compute
[157]	outputs	evade	test time	black box	read only	compute
[12]	outputs	evade	test time	white box	read only	capability
[128]	outputs	evade	test time	white box	none	statistical
[73]	none	evade	train time	none	none	compute
[162]	training run	evade audit	post hoc	none	none	none
[138]	training run	evade audit	post hoc	none	responses	statistical
[142]	outputs	evade audit	test time	black box	read only	none
[50]	outputs	evade audit	test time	black box	responses	compute
[177]	training run	evade audit	train time	white box	model	compute
[181]	outputs	inject	test time	none	channel	statistical
[151]	train data	poison	train time	none	corpus	capability
[125]	train data	poison	train time	white box	corpus	capability
[123]	train data	poison	train time	white box	corpus	capability
[131]	train data	poison	train time	black box	corpus	capability
[65]	train data	poison	train time	none	corpus	compute
[63]	train data	poison	train time	none	corpus	compute
[172]	train data	poison	train time	none	corpus	compute
[19]	train data	poison	train time	none	corpus	compute
[51]	parameters	backdoor	post hoc	white box	read only	compute
[141]	parameters	backdoor	post hoc	white box	read only	none
[137]	outputs	backdoor	post hoc	black box	read only	compute
[168]	parameters	backdoor	post hoc	white box	read only	none
[117]	parameters	backdoor	post hoc	white box	model	compute
[102]	parameters	backdoor	post hoc	black box	read only	compute
[98]	parameters	backdoor	train time	black box	model	compute
[11]	train data	backdoor	train time	none	corpus	none
[61]	parameters	backdoor	train time	white box	model	compute
[144]	parameters	backdoor	train time	none	corpus	capability
[116]	none	multiple	n/a	none	none	none
[139]	none	multiple	n/a	none	none	none
[119]	none	none	n/a	none	none	compute
[8]	none	none	n/a	none	none	none
[39]	none	none	n/a	none	none	compute
[69]	none	none	n/a	none	none	compute
[16]	none	none	train time	none	none	compute
[29]	none	none	n/a	none	none	compute
[32]	none	none	n/a	none	none	compute
[103]	none	none	n/a	none	none	compute
[18]	none	none	n/a	none	none	compute
[38]	none	none	n/a	none	none	compute
[21]	none	none	n/a	none	none	compute
[53]	train data	none	n/a	none	none	compute
[107]	none	none	n/a	none	none	compute
[133]	none	none	n/a	none	none	compute
[132]	train data	none	n/a	none	none	compute
[55]	training run	none	train time	hard label	read only	compute

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[1]	none	none	n/a	none	none	compute
[4]	train data	none	train time	none	none	capability
[31]	none	none	n/a	none	none	compute
[62]	none	none	n/a	none	none	compute
[13]	none	none	n/a	none	none	none
[80]	none	none	n/a	none	none	compute
[145]	train data	none	n/a	none	none	compute
[87]	none	none	n/a	none	none	compute
[82]	none	none	n/a	none	none	compute
[30]	none	none	train time	none	none	compute
[14]	none	none	n/a	none	none	compute
[5]	none	none	n/a	none	none	compute
[23]	none	none	n/a	none	none	compute
[20]	none	none	n/a	none	none	compute
[10]	none	none	n/a	none	none	compute
[150]	outputs	none	test time	black box	corpus	none
[96]	none	none	n/a	none	none	compute
[97]	none	none	n/a	none	none	compute
[75]	none	none	n/a	none	none	compute
[68]	none	none	n/a	none	none	compute
[126]	none	none	n/a	none	none	compute
[161]	none	none	n/a	none	none	compute
[109]	none	none	n/a	none	none	none
[112]	none	none	n/a	none	none	compute
[176]	none	none	n/a	none	none	compute
[7]	none	none	n/a	none	none	compute
[70]	outputs	none	post hoc	black box	read only	compute
[33]	none	none	n/a	none	none	none
[84]	none	none	n/a	none	none	compute
[152]	none	none	n/a	none	none	none
[3]	none	none	n/a	none	none	compute
[60]	none	none	n/a	none	none	compute
[108]	none	none	n/a	none	none	compute
[99]	none	none	n/a	none	none	compute
[49]	none	none	n/a	none	none	compute
[158]	none	none	n/a	none	none	compute
[173]	none	none	train time	none	none	compute
[57]	none	none	n/a	none	none	none
[2]	none	none	n/a	none	none	compute
[36]	none	none	n/a	none	none	compute
[171]	none	none	n/a	none	none	compute
[66]	none	none	n/a	none	none	compute
[95]	none	none	n/a	none	none	compute
[27]	none	none	n/a	none	none	compute
[124]	none	none	n/a	none	none	compute
[41]	none	none	n/a	none	none	compute
[92]	none	none	n/a	none	none	compute
[121]	none	none	n/a	none	none	compute
[28]	none	none	train time	none	none	compute
[146]	none	none	train time	none	none	compute
[17]	none	none	n/a	none	none	compute
[81]	none	none	n/a	none	none	compute
[105]	none	none	n/a	none	none	compute
[24]	none	none	n/a	none	none	compute
[100]	none	none	n/a	none	none	compute
[74]	none	none	n/a	none	none	compute
[76]	none	none	n/a	none	none	compute
[40]	none	none	train time	none	none	compute
[118]	train data	none	train time	none	none	compute

Table 5, continued

#	Target	Family	Phase	Access	Write	Frame
[180]	none	none	n/a	none	none	compute
[35]	none	none	n/a	none	none	none
[147]	train data	·	post hoc	white box	read only	capability
[78]	outputs	·	test time	none	responses	none
[148]	train data	·	post hoc	·	read only	statistical

TABLE 6: Per-work reference layer: adversary assumptions and claim shape. Same rows and same order as Table 5.

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[83]	none	none	unstated	stated	none	none	·	empirical	stated
[143]	none	none	unstated	stated	none	none	·	empirical	stated
[79]	none	none	unstated	stated	none	none	·	empirical	stated
[167]	attacker	time	poly	stated	none	none	explicit	conditional	stated
[110]	attacker	samples	exponent	stated	samples	·	explicit	universal	stated
[155]	attacker	queries	measured	stated	none	none	implicit	empirical	stated
[178]	auditor	time	unstated	stated	none	none	implicit	conditional	stated
[47]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[15]	none	none	unstated	stated	none	none	·	empirical	stated
[175]	none	none	unstated	stated	none	none	·	conditional	stated
[134]	none	none	unstated	stated	none	none	·	empirical	stated
[93]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[52]	none	none	unstated	stated	none	·	·	universal	stated
[56]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[72]	attacker	time	poly	absent	perturbation	queries	explicit	universal	stated
[120]	none	none	unstated	stated	none	·	·	existential	stated
[25]	attacker	time	measured	absent	none	none	implicit	empirical	stated
[111]	attacker	queries	exponent	stated	samples	none	explicit	conditional	stated
[130]	attacker	queries	budget	absent	privacy budget	none	explicit	universal	stated
[22]	none	none	unstated	stated	none	none	·	conditional	stated
[170]	none	none	unstated	n/a	none	none	·	empirical	stated
[106]	attacker	time	poly	stated	none	queries	explicit	conditional	stated
[163]	attacker	time	poly	stated	·	none	explicit	conditional	stated
[174]	attacker	time	poly	stated	time	none	explicit	conditional	stated
[45]	attacker	queries	budget	absent	none	none	explicit	conditional	stated
[154]	none	none	unstated	absent	none	none	·	empirical	stated
[67]	attacker	time	measured	stated	none	none	explicit	empirical	stated
[54]	attacker	queries	measured	absent	none	none	implicit	empirical	stated
[113]	attacker	algo class	class	absent	none	none	explicit	existential	stated
[46]	attacker	queries	measured	stated	none	time	implicit	empirical	stated
[165]	none	none	unstated	n/a	none	none	·	existential	stated
[77]	attacker	memory	exponent	stated	memory	none	explicit	universal	stated
[115]	none	none	unstated	absent	none	none	·	empirical	stated
[179]	attacker	time	budget	n/a	time	none	implicit	conditional	stated
[88]	attacker	time	measured	stated	none	none	implicit	empirical	stated
[64]	attacker	time	measured	stated	none	none	explicit	empirical	stated
[114]	attacker	time	unstated	absent	none	none	explicit	conditional	stated
[149]	none	none	unstated	n/a	none	none	·	empirical	stated
[164]	attacker	perturbation	budget	stated	none	none	explicit	existential	none
[48]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[153]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[37]	learner	algo class	class	n/a	perturbation	·	explicit	existential	stated
[34]	learner	time	poly	stated	perturbation	none	explicit	conditional	stated
[101]	attacker	perturbation	budget	n/a	none	none	explicit	existential	stated
[86]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[9]	learner	samples	budget	n/a	none	·	explicit	universal	stated
[122]	attacker	perturbation	budget	n/a	none	·	explicit	universal	stated
[58]	attacker	perturbation	budget	n/a	none	none	·	universal	stated
[136]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[85]	attacker	time	poly	n/a	none	none	·	universal	stated
[94]	attacker	circuit size	poly	n/a	none	none	·	existential	stated
[127]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[140]	attacker	perturbation	budget	n/a	none	none	·	conditional	stated
[135]	attacker	perturbation	budget	n/a	none	none	explicit	empirical	stated
[26]	auditor	time	exponent	stated	none	·	implicit	existential	stated
[91]	none	none	unstated	stated	none	·	·	conditional	stated
[90]	none	none	unstated	n/a	none	none	·	empirical	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[71]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[59]	attacker	perturbation	budget	n/a	none	none	explicit	conditional	stated
[6]	attacker	perturbation	budget	stated	none	none	explicit	existential	none
[42]	attacker	perturbation	budget	n/a	none	none	implicit	empirical	none
[44]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[169]	attacker	perturbation	budget	n/a	none	none	·	universal	stated
[160]	attacker	perturbation	budget	n/a	none	none	·	empirical	stated
[156]	attacker	perturbation	budget	n/a	samples	none	explicit	existential	stated
[129]	learner	samples	budget	stated	none	none	explicit	universal	stated
[43]	attacker	perturbation	budget	stated	none	time	explicit	empirical	stated
[166]	attacker	perturbation	budget	stated	none	none	explicit	existential	stated
[104]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[159]	attacker	perturbation	budget	n/a	samples	none	explicit	conditional	stated
[89]	attacker	queries	budget	stated	none	algo class	implicit	existential	none
[157]	attacker	time	measured	n/a	none	none	implicit	empirical	none
[12]	attacker	perturbation	budget	n/a	samples	none	explicit	universal	stated
[128]	learner	samples	budget	n/a	none	none	explicit	universal	stated
[73]	learner	time	poly	n/a	perturbation	none	explicit	conditional	stated
[162]	none	none	unstated	n/a	none	·	·	conditional	stated
[138]	auditor	samples	exponent	n/a	none	none	·	conditional	stated
[142]	none	none	unstated	n/a	none	none	·	existential	stated
[50]	attacker	queries	exponent	stated	none	time	explicit	existential	stated
[177]	attacker	time	exponent	n/a	none	·	implicit	existential	stated
[181]	auditor	samples	budget	n/a	none	none	explicit	conditional	stated
[151]	attacker	corruption	budget	stated	algo class	none	explicit	conditional	stated
[125]	attacker	corruption	budget	n/a	none	none	explicit	conditional	stated
[123]	attacker	corruption	budget	stated	none	·	explicit	conditional	stated
[131]	attacker	corruption	budget	stated	none	perturbation	explicit	universal	stated
[65]	learner	time	poly	n/a	corruption	·	explicit	existential	stated
[63]	learner	samples	class	stated	none	algo class	explicit	conditional	stated
[172]	learner	samples	class	stated	none	algo class	explicit	universal	stated
[19]	learner	queries	budget	stated	corruption	time	implicit	universal	stated
[51]	auditor	time	poly	stated	corruption	none	explicit	conditional	stated
[141]	none	none	unstated	n/a	none	·	·	existential	none
[137]	attacker	queries	budget	n/a	none	time	implicit	empirical	stated
[168]	none	none	unstated	n/a	none	none	·	existential	none
[117]	auditor	time	poly	stated	perturbation	none	explicit	existential	stated
[102]	auditor	time	poly	absent	corruption	samples	explicit	conditional	stated
[98]	auditor	time	poly	stated	none	none	explicit	existential	stated
[11]	none	none	unstated	n/a	none	none	·	empirical	none
[61]	auditor	time	poly	n/a	none	none	explicit	empirical	stated
[144]	attacker	corruption	measured	stated	none	·	explicit	empirical	stated
[116]	none	none	unstated	n/a	none	none	·	empirical	stated
[139]	none	none	unstated	n/a	none	none	·	empirical	stated
[119]	mechanism	memory	exponent	stated	none	·	explicit	existential	stated
[8]	none	none	unstated	n/a	none	none	·	conditional	stated
[39]	attacker	time	exponent	n/a	time	none	explicit	conditional	stated
[69]	learner	time	exponent	n/a	none	none	·	conditional	stated
[16]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[29]	auditor	algo class	class	n/a	none	none	explicit	universal	stated
[32]	attacker	circuit size	budget	n/a	circuit size	algo class	explicit	existential	stated
[103]	attacker	memory	exponent	n/a	memory	queries	explicit	universal	stated
[18]	attacker	algo class	class	n/a	none	algo class	explicit	conditional	stated
[38]	learner	time	conditional	n/a	none	algo class	explicit	conditional	stated
[21]	attacker	memory	exponent	n/a	memory	time	explicit	conditional	stated
[53]	attacker	algo class	class	stated	none	·	explicit	existential	stated
[107]	attacker	time	poly	n/a	none	none	·	conditional	stated
[133]	attacker	time	poly	n/a	none	none	·	conditional	stated
[132]	attacker	time	poly	stated	none	none	explicit	existential	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[55]	attacker	time	poly	stated	none	·	explicit	conditional	stated
[1]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[4]	mechanism	privacy budget	budget	n/a	none	none	·	empirical	stated
[31]	learner	time	poly	n/a	none	·	explicit	conditional	stated
[62]	learner	time	conditional	stated	none	algo class	explicit	conditional	stated
[13]	none	none	unstated	n/a	none	none	·	empirical	none
[80]	mechanism	memory	conditional	stated	time	none	explicit	conditional	stated
[145]	mechanism	time	budget	stated	none	·	explicit	universal	stated
[87]	learner	algo class	class	stated	none	queries	explicit	universal	stated
[82]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[30]	learner	time	conditional	n/a	none	·	explicit	conditional	stated
[14]	learner	time	conditional	n/a	none	none	explicit	conditional	stated
[5]	attacker	queries	exponent	stated	queries	none	explicit	existential	stated
[23]	attacker	time	exponent	stated	time	none	explicit	conditional	stated
[20]	attacker	circuit size	exponent	n/a	none	none	explicit	existential	stated
[10]	attacker	time	conditional	n/a	none	none	explicit	conditional	stated
[150]	none	none	unstated	n/a	none	none	·	empirical	stated
[96]	learner	circuit size	conditional	n/a	none	none	explicit	universal	stated
[97]	learner	algo class	class	n/a	none	·	explicit	universal	stated
[75]	learner	samples	exponent	stated	none	algo class	explicit	conditional	stated
[68]	learner	samples	exponent	absent	none	algo class	explicit	conditional	stated
[126]	learner	time	poly	stated	none	·	explicit	conditional	stated
[161]	learner	time	poly	stated	perturbation	·	explicit	conditional	stated
[109]	none	none	unstated	n/a	none	·	·	universal	stated
[112]	learner	time	poly	n/a	none	algo class	explicit	existential	stated
[176]	mechanism	algo class	class	stated	none	time	implicit	conditional	stated
[7]	learner	algo class	class	n/a	none	samples	explicit	universal	stated
[70]	auditor	algo class	class	n/a	privacy budget	samples	explicit	existential	stated
[33]	none	none	unstated	n/a	none	none	·	conditional	stated
[84]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[152]	none	none	unstated	n/a	none	none	·	universal	stated
[3]	learner	algo class	class	n/a	none	time	explicit	conditional	stated
[60]	learner	algo class	class	n/a	none	queries	explicit	universal	stated
[108]	learner	time	poly	n/a	none	none	explicit	conditional	stated
[99]	attacker	time	poly	n/a	privacy budget	none	explicit	conditional	stated
[49]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[158]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[173]	learner	algo class	class	n/a	none	·	explicit	conditional	stated
[57]	none	none	unstated	n/a	none	none	·	conditional	stated
[2]	learner	algo class	class	n/a	none	samples	explicit	existential	stated
[36]	attacker	queries	exponent	n/a	queries	none	explicit	universal	stated
[171]	attacker	algo class	class	stated	algo class	none	explicit	conditional	stated
[66]	learner	samples	exponent	stated	none	algo class	explicit	conditional	stated
[95]	learner	parameters	budget	stated	none	time	explicit	existential	stated
[27]	attacker	time	exponent	n/a	time	none	explicit	conditional	stated
[124]	attacker	time	exponent	n/a	none	none	explicit	conditional	stated
[41]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[92]	attacker	circuit size	quasi-poly	stated	none	none	·	conditional	stated
[121]	learner	time	exponent	n/a	none	success prob	explicit	conditional	stated
[28]	learner	time	quasi-poly	n/a	none	algo class	explicit	existential	stated
[146]	learner	memory	budget	n/a	none	none	explicit	universal	stated
[17]	learner	time	conditional	stated	corruption	algo class	explicit	conditional	stated
[81]	learner	samples	exponent	stated	none	time	explicit	universal	stated
[105]	auditor	algo class	class	n/a	none	·	explicit	conditional	stated
[24]	mechanism	time	poly	n/a	time	none	explicit	conditional	stated
[100]	attacker	circuit depth	exponent	n/a	none	queries	explicit	existential	stated
[74]	attacker	time	poly	n/a	privacy budget	none	explicit	universal	stated
[76]	attacker	memory	exponent	stated	memory	·	explicit	universal	stated
[40]	learner	parameters	budget	n/a	none	none	explicit	conditional	stated

Table 6, continued

#	Party	Unit	Tightness	Baseline	2nd party	2nd notion	Assessed	Quantifier	Antecedent
[118]	learner	algo class	class	n/a	none	none	explicit	universal	stated
[180]	learner	time	poly	stated	none	samples	explicit	existential	stated
[35]	none	none	unstated	n/a	none	none	·	empirical	stated
[147]	mechanism	privacy budget	budget	stated	none	none	implicit	universal	stated
[78]	none	none	unstated	n/a	none	none	·	empirical	stated
[148]	attacker	samples	measured	stated	none	none	implicit	empirical	stated

References

[1] D. Abram, A. Beimel, Y. Ishai, E. Kushilevitz, and V. Narayanan, "Cryptography from planted graphs: Security with logarithmic-size messages," in *IACR ePrint/*, 2023.

[2] E. Abbe, E. Boix-Adserà, and T. Misiakiewicz, "The merged-staircase property - a necessary and nearly sufficient condition for sgd learning of sparse functions on two-layer neural networks," arXiv:2202.08658, 2022.

[3] ——, "Sgd learning on neural networks - leap complexity and saddle-to-saddle dynamics," arXiv:2302.11055, 2023.

[4] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *CCS*, 2016.

[5] A. Afshar, G. Couteau, M. Mahmoody, and E. Sadeghi, "Fine-grained non-interactive key-exchange: Constructions and lower bounds," in *venue unconfirmed from PDF*, 2023.

[6] A. Athalye, N. Carlini, and D. Wagner, "Obfuscated gradients give a false sense of security — circumventing defenses to adversarial examples," in *ICML*, 2018.

[7] G. B. Arous, R. Gheissari, and A. Jagannath, "Online stochastic gradient descent on non-convex losses from high-dimensional inference," arXiv:2003.10409, 2021.

[8] D. Alvarez-Melis, V. Garg, and A. T. Kalai, "Are gans overkill for nlp?" in *NeurIPS — stated in the PDF: "36th Conference on Neural Information Processing Systems*, 2022.

[9] I. Attias and S. Hanneke, "Adversarially robust pac learnability of real-valued functions," arXiv:2206.12977, 2023.

[10] J. Alman, Y. Huang, and K. Yeo, "Fine-grained complexity in a world without cryptography," in *Eurocrypt*, 2025.

[11] G. Abad, M. Krček, S. Koffas, B. Tajalli, M. Arazzi, R. Riaño, X. Xu, Z. Liu, A. Nocera, and S. Picek, "Sok: The last line of defense — on backdoor defense evaluation," arXiv:2511.13143, 2025.

[12] H. Ashtiani, V. Pathak, and R. Urner, "Simplifying adversarially robust pac learning with tolerance," arXiv:2502.07232, 2025.

[13] D. Arp, E. Quiring, F. Pendlebury, A. Warnecke, F. Pierazzi, C. Wressnegger, L. Cavallaro, and K. Rieck, "Dos and don'ts of machine learning in computer security," in *USENIX Security*, 2022.

[14] J. Alman and Z. Song, "The fine-grained complexity of gradient computation for training large language models," arXiv:2402.04497, 2024.

[15] M. Aerni, J. Zhang, and F. Tramèr, "Evaluations of machine learning privacy defenses are misleading," in *ACM CCS*, 2024.

[16] A. S. Bandeira, A. E. Alaoui, S. B. Hopkins, T. Schramm, A. S. Wein, and I. Zadik, "The franz–parisi criterion and computational trade-offs in high dimensional statistics," arXiv:2205.09727, 2022.

[17] M. Brennan and G. Bresler, "Reducibility and statistical-computational gaps from secret leakage," arXiv:2005.08099, 2020.

[18] M. Brennan, G. Bresler, S. B. Hopkins, J. Li, and T. Schramm, "Statistical query algorithms and low-degree tests are almost equivalent," in *COLT*, 2021.

[19] M.-F. Balcan, A. Blum, S. Hanneke, and D. Sharma, "Robustly-reliable learners under poisoning attacks," arXiv:2203.04160, 2022.

[20] C. Brzuska and G. Couteau, "On building fine-grained one-way functions from strong average-case hardness," in *EUROCRYPT*, 2022.

[21] C. Brzuska, G. Couteau, C. Egger, and W. Quach, "On bounded storage key agreement and one-way functions," in *Theory of Cryptography Conference*, 2024.

[22] B. Balle, G. Cherubin, and J. Hayes, "Reconstructing training data with informed adversaries," in *IEEE S&P*, 2022.

[23] B. Bauer, G. Couteau, and E. Sadeghi, "Fine-grained non-interactive key exchange, revisited," in *venue unconfirmed from PDF*, 2024.

[24] M. Bun, Y.-H. Chen, and S. Vadhan, "Separating computational and statistical differential privacy in the client-server model," in *venue unconfirmed from PDF*, 2016.

[25] M. Balunović, D. I. Dimitrov, N. Jovanović, and M. Vechev, "Lamp: Extracting text from gradients with language model priors," in *Conference on Neural Information Processing Systems*, 2022.

[26] S. Ball, G. Gluch, S. Goldwasser, F. Kreuter, O. Reingold, and G. N. Rothblum, "On the impossibility of separating intelligence from judgment: The computational intractability of filtering for ai alignment," arXiv:2507.07341, 2025.

[27] M. Ball, J. Garay, P. Hall, A. Kiayias, and G. Panagiotakos, "Towards permissionless consensus in the standard model via fine-grained complexity," in *venue unconfirmed from PDF*, 2024.

[28] R.-D. Buhai, J.-T. Hsieh, A. Jain, and P. K. Kothari, "The quasi-polynomial low-degree conjecture is false," in *FOCS*, 2025.

[29] B. Barak, S. B. Hopkins, J. Kelner, P. K. Kothari, A. Moitra, and A. Potechin, "A nearly tight sum-of-squares lower bound for the planted clique problem," in *FOCS*, 2016.

[30] A. Backurs, P. Indyk, and L. Schmidt, "On the fine-grained complexity of empirical risk minimization: Kernel methods and neural networks," arXiv:1704.02958, 2017.

[31] G. Bresler and T. Jiang, "Detection-recovery and detection-refutation gaps via reductions from planted clique," in *COLT. The ingested PDF states its venue: Proceedings of Machine Learning Research*, 2023.

[32] A. Bogdanov, C. Jones, A. Rosen, and I. Zadik, "Low-degree security of the planted random subgraph problem," arXiv:2409.16227, 2024.

[33] H. Brown, K. Lee, F. Mireshghallah, R. Shokri, and F. Tramèr, "What does it mean for a language model to preserve privacy?" arXiv:2202.05520, 2022.

[34] S. Bubeck, Y. T. Lee, E. Price, and I. Razenshteyn, "Adversarial examples from cryptographic pseudo-random generators," arXiv:1811.06418, 2018.

[35] K. Balan, R. Learney, and T. Wood, "A framework for cryptographic verifiability of end-to-end ai pipelines," in *IWSPA '25*, 2025.

[36] B. Barak and M. Mahmoody-Ghidary, "Merkle puzzles are optimal — an $o(n^2)$-query attack on any key exchange from a random oracle," in *CRYPTO — stated in the PDF's own footer: "S. Halevi*, 2009.

[37] S. Bubeck, E. Price, and I. Razenshteyn, "Adversarial examples from computational constraints," arXiv:1805.10204, 2019.

[38] J. Bruna, O. Regev, M. J. Song, and Y. Tang, "Continuous lwe," in *STOC*, 2021.

[39] M. Ball, A. Rosen, M. Sabin, and P. N. Vasudevan, "Average-case fine-grained hardness," in *venue unconfirmed from PDF*, 2017.

[40] S. Bubeck and M. Sellke, "A universal law of robustness via isoperimetry," arXiv:2105.12806, 2021.

[41] B. Barak and A. Wigderson, "Public-key cryptography from different assumptions," in *IACR ePrint/335*, 2008.

[42] N. Carlini, A. Athalye, N. Papernot, W. Brendel, J. Rauber, D. Tsipras, I. Goodfellow, A. Mądry, and A. Kurakin, "On evaluating adversarial robustness," arXiv:1902.06705, 2019.

[43] F. Croce, M. Andriushchenko, V. Sehwag, E. Debenedetti, N. Flammarion, M. Chiang, P. Mittal, and M. Hein, "Robustbench — a standardized adversarial robustness benchmark," in *NeurIPS Track on Datasets and Benchmarks*, 2021.

[44] D. Cullina, A. N. Bhagoji, and P. Mittal, "Pac-learning in the presence of evasion adversaries," arXiv:1806.01471, 2018.

[45] V. Chandrasekaran, K. Chaudhuri, I. Giacomelli, S. Jha, and S. Yan, "Exploring connections between active learning and model extraction," in *USENIX Security*, 2019.

[46] I. A. Canales-Martínez, J. Chávez-Saab, A. Hambitzer, F. Rodríguez-Henríquez, N. Satpute, and A. Shamir, "Polynomial time cryptanalytic extraction of neural network models," in *EUROCRYPT*, 2024.

[47] N. Carlini, S. Chien, M. Nasr, S. Song, A. Terzis, and F. Tramèr, "Membership inference attacks from first principles," in *IEEE S&P*, 2022.

[48] F. Croce, S. Gowal, T. Brunner, E. Shelhamer, M. Hein, and T. Cemgil, "Evaluating the adversarial robustness of adaptive test-time defenses," in *ICML*, 2022.

[49] A. Carpentier, S. M. Giancola, C. Giraud, and N. Verzelen, "Low-degree lower bounds via almost orthonormal bases," arXiv:2509.09353, 2025.

[50] M. Christ, S. Gunn, and O. Zamir, "Undetectable watermarks for language models," arXiv:2306.09194, 2023.

[51] P. Christiano, J. Hilton, V. Lecomte, and M. Xu, "Backdoor defense, learnability, and obfuscation," arXiv:2409.03077, 2025.

[52] R. Cummings, S. Hod, J. Sarathy, and M. Swanberg, "Attaxonomy: Unpacking differential privacy guarantees against practical adversaries," arXiv:2405.01716, 2024.

[53] K. Chaudhuri, J. Imola, and A. Machanavajjhala, "Capacity bounded differential privacy," arXiv:1907.02159, 2019.

[54] N. Carlini, M. Jagielski, and I. Mironov, "Cryptanalytic extraction of neural network models," in *CRYPTO*, 2020.

[55] R. Canetti and A. Karchmer, "Covert learning — how to learn with an untrusted intermediary," in *PDF in 'raw/' is dated 17 September and names no venue. The TCC attribution carried on this page is unconfirmed from the primary source*, 2021.

[56] E. Cohen, H. Kaplan, Y. Mansour, S. Moran, K. Nissim, U. Stemmer, and E. Tsfadia, "Data reconstruction: When you see it and when you don't," in *venue unconfirmed from PDF*, 2024.

[57] S. Chakraborty, J. Qiu, H. Yuan, A. Koppel, D. Manocha, F. Huang, A. S. Bedi, and M. Wang, "Maxmin-rlhf — alignment with diverse human preferences (impossibility of single-reward rlhf)," arXiv:2402.08925, 2024.

[58] J. Cohen, E. Rosenfeld, and J. Z. Kolter, "Certified adversarial robustness via randomized smoothing," in *ICML*, 2019.

[59] Z. Cheng, J. Yang, W. Dai, and J. Sun, "Provable defense framework for llm jailbreaks via noise-augmented alignment (naat + css)," arXiv:2602.01587, 2026.

[60] A. Das, S. Gollapudi, R. Kumar, and R. Panigrahy, "On the learnability of deep random networks," arXiv:1904.03866, 2019.

[61] A. Draguns, A. Gritsevskiy, S. R. Motwani, C. Rogers-Smith, J. Ladish, and C. S. de Witt, "Unelicitable backdoors in language models via cryptographic transformer circuits," arXiv:2406.02619, 2024.

[62] J. Ding, Y. Hua, L. Slot, and D. Steurer, "Low degree conjecture implies sharp computational thresholds in stochastic block model," arXiv:2502.15024, 2025.

[63] I. Diakonikolas, G. Iakovidis, D. M. Kane, and L. Ren, "Algorithms and sq lower bounds for robustly learning real-valued multi-index models," arXiv:2505.21475, 2025.

[64] H. Deng, Y. Jiang, G. Yu, Q. Wang, X. Wang, B. Ma, W. Ni, and R. P. Liu, "Polo: Proof-of-learning and proof-of-ownership at once with chained watermarking," arXiv:2505.12296, 2026.

[65] I. Diakonikolas, G. Kamath, D. M. Kane, J. Li, A. Moitra, and A. Stewart, "Robust estimators in high dimensions without the computational intractability," arXiv:1604.06443, 2016.

[66] L. Defilippis, F. Krzakala, B. Loureiro, and A. Maillard, "A noise sensitivity exponent controls large statistical-to-computational gaps in single- and multi-index models," arXiv:2603.17896, 2026.

[67] A. Dziedzic, M. A. Kaleem, Y. S. Lu, and N. Papernot, "Increasing the cost of model extraction with calibrated proof of work," in *ICLR*, 2022.

[68] A. Damian, J. D. Lee, and J. Bruna, "The generative leap - sharp sample complexity for efficiently learning gaussian multi-index models," arXiv:2506.05500, 2025.

[69] M. Dalirrooyfard, A. Lincoln, B. Saha, and V. V. Williams, "Average-case hardness of parity problems: Orthogonal vectors, k-sum and more," arXiv:2503.21951, 2025.

[70] C. Domingo-Enrich and Y. Mroueh, "Auditing differential privacy in high dimensions with the kernel quantum rényi divergence," arXiv:2205.13941, 2022.

[71] D. I. Diochnos, S. Mahloujifar, and M. Mahmoody, "Lower bounds for adversarially robust pac learning," arXiv:1906.05815, 2019.

[72] I. Dinur and K. Nissim, "Revealing information while preserving privacy," in *venue unconfirmed from PDF*, 2003.

[73] A. Degwekar, P. Nakkiran, and V. Vaikuntanathan, "Computational limitations in robust classification and win-win results," in *COLT*, 2019.

[74] D. Desfontaines and B. Pejó, "Sok: Differential privacies," in *Proceedings on Privacy Enhancing Technologies*, 2020.

[75] A. Damian, L. Pillaud-Vivien, J. D. Lee, and J. Bruna, "Computational-statistical gaps in gaussian single-index models," arXiv:2403.05529, 2024.

[76] Y. Dodis, W. Quach, and D. Wichs, "Speak much, remember little: Cryptography in the bounded storage model, revisited," in *venue unconfirmed from PDF*, 2021.

[77] ——, "Authentication in the bounded storage model," in *venue unconfirmed from PDF*, 2022.

[78] E. Debenedetti, I. Shumailov, T. Fan, J. Hayes, N. Carlini, D. Fabian, C. Kern, C. Shi, A. Terzis, and F. Tramèr, "Defeating prompt injections by design (camel)," arXiv:2503.18813, 2025.

[79] M. Duan, A. Suri, N. Mireshghallah, S. Min, W. Shi, L. Zettlemoyer, Y. Tsvetkov, Y. Choi, D. Evans, and H. Hajishirzi, "Do membership inference attacks work on large language models?" in *COLM*, 2024.

[80] I. Dinur, U. Stemmer, D. P. Woodruff, and S. Zhou, "On differential privacy and adaptive data analysis with bounded space," in *venue unconfirmed from PDF*, 2023.

[81] Y. Dandi, E. Troiani, L. Arnaboldi, L. Pesce, L. Zdeborová, and F. Krzakala, "The benefits of reusing batches for gradient descent in two-layer networks - breaking the curse of information and leap exponents," arXiv:2402.03220, 2024.

[82] A. Degwekar, V. Vaikuntanathan, and P. N. Vasudevan, "Fine-grained cryptography," in *venue unconfirmed from PDF*, 2016.

[83] D. Das, J. Zhang, and F. Tramèr, "Blind baselines beat membership inference attacks for foundation models," in *DATA-FM workshop @ ICLR*, 2025.

[84] B. Even, C. Giraud, and N. Verzelen, "Computational lower bounds in latent models: Clustering, sparse-clustering, biclustering," arXiv:2506.13647, 2025.

[85] O. Etesami, S. Mahloujifar, and M. Mahmoody, "Computational concentration of measure — optimal bounds, reductions, and more," arXiv:1907.05401, 2020.

[86] A. Fawzi, H. Fawzi, and O. Fawzi, "Adversarial vulnerability for any classifier," in *NeurIPS*, 2018.

[87] V. Feldman, E. Grigorescu, L. Reyzin, S. S. Vempala, and Y. Xiao, "Statistical algorithms and a lower bound for detecting planted cliques," in *JACM*, 2017.

[88] C. Fang, H. Jia, A. Thudi, M. Yaghini, C. A. Choquette-Choo, N. Dullerud, V. Chandrasekaran, and N. Papernot, "Proof-of-learning is currently more broken than you think," arXiv:2208.03567, 2023.

[89] L. Fenaux and F. Kerschbaum, "Sok: Analyzing adversarial examples — a framework to study adversary knowledge," arXiv:2402.14937, 2024.

[90] M. A. Ferrag, N. Tihanyi, D. Hamouda, L. Maglaras, A. Lakas, and M. Debbah, "From prompt injections to protocol exploits: Threats in llm-powered ai agents workflows (imtm origin)," arXiv:2506.23260, 2025.

[91] S. S. Ghosal, S. Chakraborty, V. Singh, T. Guan, M. Wang, A. Velasquez, A. Beirami, F. Huang, D. Manocha, and A. S. Bedi, "Immune — improving safety against jailbreaks in multi-modal llms via inference-time alignment," arXiv:2411.18688, 2025.

[92] R. Ghosal, I. M. Hair, A. Jain, and A. Sahai, "Using the planted clique conjecture for cryptography: Public-key encryption from planted clique and noisy k-lin over expanders," in *STOC*, 2025.

[93] B. Ghazi, R. Ilango, P. Kamath, R. Kumar, and P. Manurangsi, "Towards separating computational and statistical differential privacy," arXiv:2301.00104, 2023.

[94] S. Garg, S. Jha, S. Mahloujifar, and M. Mahmoody, "Adversarially robust learning could leverage computational hardness," arXiv:1905.11564, 2020.

[95] S. Garg, S. Jha, S. Mahloujifar, M. Mahmoody, and M. Wang, "Overparameterization from computational constraints," arXiv:2208.12926, 2022.

[96] D. Gamarnik, A. Jagannath, and A. S. Wein, "Circuit lower bounds for the p-spin optimization problem," arXiv:2109.01342, 2022.

[97] ——, "Hardness of random optimization problems for boolean circuits, low-degree polynomials, and langevin dynamics," in *FOCS*, 2022.

[98] S. Goldwasser, M. P. Kim, V. Vaikuntanathan, and O. Zamir, "Planting undetectable backdoors in machine learning models," arXiv:2204.06974, 2022.

[99] A. Groce, J. Katz, and A. Yerukhimovich, "Limits of computational differential privacy in the client/server setting," in *Theory of Cryptography Conference*, 2011.

[100] J. Guan and H. Montgomery, "On sequential functions and fine-grained cryptography," in *CRYPTO*, 2024.

[101] J. Gilmer, L. Metz, F. Faghri, S. S. Schoenholz, M. Raghu, M. Wattenberg, and I. Goodfellow, "The relationship between high-dimensional geometry and adversarial examples (adversarial spheres)," arXiv:1801.02774, 2018.

[102] S. Goldwasser, J. Shafer, N. Vafa, and V. Vaikuntanathan, "Oblivious defense in ml models: Backdoor removal without detection," arXiv:2411.03279, 2025.

[103] J. Guan and M. Zhandry, "Simple schemes in the bounded storage model," in *venue unconfirmed from PDF*, 2019.

[104] J. Hu, Y. Dong, Z. Ding, and X. Huang, "Enhancing robustness of llm-driven multi-agent systems through randomized smoothing," arXiv:2507.04105, 2025.

[105] J.-T. Hsieh, D. M. Kane, P. K. Kothari, J. Li, S. Mohanty, and S. Tiegel, "Rigorous implications of the low-degree heuristic," in *STOC*, 2026.

[106] I. Haitner, N. Mazor, J. Silbak, and E. Tsfadia, "On the complexity of two-party differential privacy," arXiv:2108.07664, 2022.

[107] I. Haitner, N. Mazor, J. Silbak, E. Tsfadia, and C. Yan, "Computationally differentially private inner-product protocols imply oblivious transfer," arXiv:2502.15629, 2025.

[108] S. Hirahara and M. Nanashima, "Learning in pessiland via inductive inference," in *Electronic Colloquium on Computational Complexity*, 2023.

[109] S. Hirahara and N. Shimizu, "Planted clique conjectures are equivalent," in *ECCC Report No. 58*, 2024.

[110] M. Haghifam, A. Smith, and J. Ullman, "The sample complexity of membership inference and privacy auditing," in *venue unconfirmed from PDF*, 2025.

[111] M. Hardt and J. Ullman, "Preventing false discovery in interactive data analysis is hard," arXiv:1408.1655, 2014.

[112] J. Holmgren and A. S. Wein, "Counterexamples to the low-degree conjecture," in *ITCS*, 2021.

[113] M. Jagielski, N. Carlini, D. Berthelot, A. Kurakin, and N. Papernot, "High accuracy and high fidelity extraction of neural networks," in *USENIX Security*, 2020.

[114] H. Jia, M. Yaghini, C. A. Choquette-Choo, N. Dullerud, A. Thudi, V. Chandrasekaran, and N. Papernot, "Proof-of-learning: Definitions and practice," in *IEEE S&P*, 2021.

[115] Z. Kong, A. R. Chowdhury, and K. Chaudhuri, "Can membership inferencing be refuted?" arXiv:2303.03648, 2023.

[116] R. Ko, J. Jeong, S. Zheng, C. Xiao, T.-W. Kim, M. Onizuka, and W.-Y. Shin, "Seven security challenges that must be solved in cross-domain multi-agent llm systems," arXiv:2505.23847, 2025.

[117] A. Kalavasis, A. Karbasi, A. Oikonomou, K. Sotiraki, G. Velegkas, and M. Zampetakis, "Injecting undetectable backdoors in obfuscated neural networks and language models," arXiv:2406.05660, 2024.

[118] S. P. Kasiviswanathan, H. K. Lee, K. Nissim, S. Raskhodnikova, and A. Smith, "What can we learn privately?" in *FOCS*, 2008.

[119] H. Kaplan, Y. Mansour, K. Nissim, and U. Stemmer, "Separating adaptive streaming from oblivious streaming," arXiv:2101.10836, 2021.

[120] ——, "Bayesian perspective on memorization and reconstruction," in *venue unconfirmed from PDF*, 2025.

[121] D. Kühnemann, A. Polak, and A. Rosen, "The planted orthogonal vectors problem," arXiv:2505.00206, 2025.

[122] M. Lecuyer, V. Atlidakis, R. Geambasu, D. Hsu, and S. Jana, "Certified robustness to adversarial examples with differential privacy (pixeldp)," arXiv:1802.03471, 2019.

[123] S. Liu, A. C. Cullen, P. Montague, S. Erfani, and B. I. P. Rubinstein, "Multi-level certified defense against poisoning attacks in offline reinforcement learning," arXiv:2505.20621, 2025.

[124] R. LaVigne, A. Lincoln, and V. V. Williams, "Public-key cryptography in the fine-grained setting," in *CRYPTO*, 2019.

[125] Y. Lai, X. Xue, L. Shen, Y. Wu, G. Li, S. Guo, K. Zhou, and B. Xiao, "Provably robust adaptation for language-empowered foundation models (lefcert, lefcert-l, lefcert-c)," arXiv:2510.08659, 2025.

[126] S. Li, I. Zadik, and M. Zampetakis, "On the hardness of learning one hidden layer neural networks," arXiv:2410.03477, 2024.

[127] S. Mahloujifar, D. I. Diochnos, and M. Mahmoody, "The curse of concentration in robust learning — evasion and poisoning attacks from concentration of measure," in *AAAI*, 2019.

[128] O. Montasser, S. Hanneke, and N. Srebro, "Vc classes are adversarially robustly learnable, but only improperly," in *COLT*, 2019.

[129] ——, "Adversarially robust learning: A generic minimax optimal learner and characterization," arXiv:2209.07369, 2022.

[130] P. Mukherjee and S. Lokam, "On the query complexity of training data reconstruction in private learning," arXiv:2303.16372, 2024.

[131] S. Mahloujifar, M. Mahmoody, and A. Mohammed, "Universal multi-party poisoning attacks," in *ICML*, 2019.

[132] I. Mironov, O. Pandey, O. Reingold, and S. Vadhan, "Computational differential privacy," in *CRYPTO — not stated in this PDF; attribution taken from the reference lists of DP20_SoKDifferentialPrivacies*, 2009.

[133] F. Meisingseth and C. Rechberger, "Relating definitions of computational differential privacy in wider parameter regimes," in *venue unconfirmed from PDF*, 2025.

[134] M. Meeus, I. Shilov, S. Jain, M. Faysse, M. Rei, and Y.-A. de Montjoye, "Sok: Membership inference attacks on llms are rushing nowhere (and how to fix it)," in *IEEE SaTML*, 2025.

[135] S. Mahloujifar, X. Zhang, M. Mahmoody, and D. Evans, "Empirically measuring concentration — fundamental limits on intrinsic robustness," in *NeurIPS*, 2019.

[136] S. Mahloujifar, "A complexity theoretic approach to adversarial machine learning," in *Research Statement / PhD Thesis Overview*, 2019.

[137] A. T. Ngo, A. Chattopadhyay, and S. Maitra, "Cryptographic backdoor for neural networks: Boon and bane," arXiv:2509.20714, 2025.

[138] M. Ngo and M. P. Kim, "Publicly-verifiable certificates for statistical algorithms," arXiv:2607.15528, 2026.

[139] N. Papernot, P. McDaniel, A. Sinha, and M. P. Wellman, "Sok: Security and privacy in machine learning," in *IEEE EuroS&P*, 2018.

[140] A. Pal, J. Sulam, and R. Vidal, "Adversarial examples might be avoidable — the role of data concentration in adversarial robustness," in *NeurIPS*, 2023.

[141] H. Qiu, H. Ma, Z. Zhang, A. Abuadbba, W. Kang, A. Fu, and Y. Gao, "Towards a critical evaluation of robustness for deep learning backdoor countermeasures," arXiv:2204.06273, 2022.

[142] A. Rao, M. Choudhury, and S. Aditya, "Jailbreak paradox: The achilles' heel of llms," arXiv:2406.12702, 2024.

[143] S. Rezaei and X. Liu, "On the difficulty of membership inference attacks," arXiv:2005.13702, 2021.

[144] J. Rando and F. Tramèr, "Universal jailbreak backdoors from poisoned human feedback," arXiv:2311.14455, 2023.

[145] Z. Ratliff and S. Vadhan, "A framework for differential privacy against timing attacks," arXiv:2409.05623, 2024.

[146] R. Raz, "Fast learning requires good memory: A time-space lower bound for parity learning," arXiv:1602.05161, 2016.

[147] M. Swanberg, M. S. M. S. Annamalai, J. Hayes, B. Balle, and A. Smith, "A unified framework for adversary-aware differential privacy bounds," arXiv:2507.08158, 2026.

[148] A. Suri and D. Evans, "Formalizing and estimating distribution inference risks," arXiv:2109.06024, 2022.

[149] M. Suliman, A. Halimi, S. R. Kadhe, N. Baracaldo, and D. Leith, "Towards a re-evaluation of data forging attacks in practice," arXiv:2411.05658, 2025.

[150] V. Siu, J. He, K. Montgomery, Z. Wang, N. Gong, C. Wang, and D. Song, "A framework for formalizing llm agent security," arXiv:2603.19469, 2026.

[151] J. Steinhardt, P. W. Koh, and P. Liang, "Certified defenses for data poisoning attacks," arXiv:1706.03691, 2017.

[152] S. Sonoda, K. Kasaura, Y. Mizuno, K. Tsukamoto, and N. Onda, "Lean formalization of generalization error bound by rademacher complexity and dudley's entropy integral," arXiv:2503.19605, 2026.

[153] F. Shpilevskiy, S. Lyu, K. D. Dvijotham, M. Lécuyer, and P.-A. Noël, "Adaptive diffusion denoised smoothing (adds)," arXiv:2507.08163, 2025.

[154] A. Shafran, I. Shumailov, M. A. Erdogdu, and N. Papernot, "Beyond labeling oracles: What does it mean to steal ml models?" arXiv:2310.01959, 2024.

[155] R. Shokri, M. Stronati, C. Song, and V. Shmatikov, "Membership inference attacks against machine learning models," arXiv:1610.05820, 2017.

[156] L. Schmidt, S. Santurkar, D. Tsipras, K. Talwar, and A. Mądry, "Adversarially robust generalization requires more data," arXiv:1804.11285, 2018.

[157] F. Suya, A. Suri, T. Zhang, J. Hong, Y. Tian, and D. Evans, "Sok: Pitfalls in evaluating black-box attacks," arXiv:2310.17534, 2024.

[158] T. Schramm and A. S. Wein, "Computational barriers to estimation from low-degree polynomials," in *Annals of Statistics*, 2022.

[159] E. Seferis, C. Wu, S. Kollias, S. Bensalem, and C.-H. Cheng, "Randomized smoothing meets vision-language models (rs-vlm)," arXiv:2509.16088, 2025.

[160] H. Salman, G. Yang, J. Li, P. Zhang, H. Zhang, I. Razenshteyn, and S. Bubeck, "Provably robust deep learning via adversarially trained smoothed classifiers (smoothadv)," in *NeurIPS*, 2019.

[161] M. J. Song, I. Zadik, and J. Bruna, "On the cryptographic hardness of learning single periodic neurons," arXiv:2106.10744, 2021.

[162] Y. Shavit, "What does it take to catch a chinchilla? verifying rules on large-scale neural network training via compute monitoring," arXiv:2303.11341, 2023.

[163] M. J. Song, "Cryptographic hardness of score estimation," arXiv:2404.03272, 2024.

[164] F. Tramèr, N. Carlini, W. Brendel, and A. Mądry, "On adaptive attacks to adversarial example defenses," in *NeurIPS*, 2020.

[165] A. Thudi, H. Jia, I. Shumailov, and N. Papernot, "On the necessity of auditable algorithmic definitions for machine unlearning," in *USENIX Security*, 2022.

[166] D. Tsipras, S. Santurkar, L. Engstrom, A. Turner, and A. Mądry, "Robustness may be at odds with accuracy," arXiv:1805.12152, 2019.

[167] F. Villa, F. B. Durak, T. Kohno, T. Maharramli, and F. Roesner, "Extending the formalism and theoretical foundations of cryptography to ai," arXiv:2603.02590, 2026.

[168] A. K. Veldanda and S. Garg, "On evaluating neural network backdoor defenses," arXiv:2010.12186, 2020.

[169] E. Wong and J. Z. Kolter, "Provable defenses against adversarial examples via the convex outer adversarial polytope," in *ICML*, 2018.

[170] R. Wen, Y. Liu, M. Backes, and Y. Zhang, "Sok: Data reconstruction attacks against machine learning models: Definition, metrics, and benchmark," arXiv:2506.07888, 2025.

[171] Y. Wang, C. Su, J. Pan, and C. Xu, "Fine-grained non-interactive key-exchange without idealized assumptions, and lower bounds," in *Crypto*, 2024.

[172] P. Wang, N. Zarifis, I. Diakonikolas, and J. Diakonikolas, "Sample and computationally efficient robust learning of gaussian single-index models," arXiv:2411.05708, 2024.

[173] A. S. Wein, "Computational complexity of statistics: New insights from low-degree polynomials," in *Survey. The ingested PDF is arXiv:2506.10748v1*, 2025.

[174] H. Xiao, G. E. Suh, and S. Devadas, "Formal privacy proof of data encoding: The possibility and impossibility of learnable obfuscation," in *CCS*, 2024.

[175] S. Yeom, I. Giacomelli, M. Fredrikson, and S. Jha, "Privacy risk in machine learning: Analyzing the connection to overfitting," in *CSF*, 2018.

[176] J. Yu, Y. He, A. Goyal, and S. Arora, "On the impossibility of retrain equivalence in machine unlearning," arXiv:2510.16629, 2025.

[177] B. Zhang, Z. Chen, C. Shen, and J. Li, "Verification of machine unlearning is fragile," in *ICML*, 2024.

[178] J. Zhang, D. Das, G. Kamath, and F. Tramèr, "Position: Membership inference attacks cannot prove that a model was trained on your data," arXiv:2409.19798, 2025.

[179] Z. Zhao, Z. Fang, X. Wang, X. Chen, H. Su, H. Xiao, and Y. Zhou, "Proof-of-learning with incentive security," arXiv:2404.09005, 2026.

[180] I. Zadik, M. J. Song, A. S. Wein, and J. Bruna, "Lattice-based methods surpass sum-of-squares in clustering," in *COLT*, 2022.

[181] K. Zhu, X. Yang, J. Wang, W. Guo, and W. Y. Wang, "Melon: Provable defense against indirect prompt injection attacks in ai agents," arXiv:2502.05174, 2025.